Mid SoL
DS(CS)

Q1(a) $\left\lceil \frac{677}{38} \right\rceil$ (2 Marks)

$\lceil 17.817 \rceil$

$\boxed{18}$ Ans

(3 Marks)

Q1(b) (i) One to one Every Mobile No. has Unique

(ii) Onto it is not necessary that b/c many students have the same grade

(iii) Onto it is not necessary that only one student lives in a particular area.

Q1(c) (i) I am not hungry

(ii) I am tired and I am not hungry

(iii) Nor I am tired or Neither I am hungry

(iv) I am tired implies I am hungry

(v) I am tired NOR I am hungry

(d) $p \uparrow q \uparrow r$ (5 Marks)

p	q	r	$p \uparrow q$	$p \uparrow q \uparrow r$
T	T	T	F	T
T	T	F	F	T
T	F	T	T	F
T	F	F	T	T
F	T	T	T	F
F	T	F	T	T
F	F	T	T	F
F	F	F	T	T

Q2(a) (2 Marks)

~~let~~

$$|A| = \left|\frac{500}{2}\right| = 250$$

$$|B| = \left|\frac{500}{5}\right| = 100$$

$$|A \cap B| = \left|\frac{500}{10}\right| = 50$$

$$|P \cup Q| = |P| + |Q| - |P \cap Q| \quad \text{—— } (*)$$

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$= 250 + 100 - 50$$

$$= 300$$

Not divisible by 2 or 5 $= 500 - 300$

$$\boxed{= 200}$$

Q2(b) $A=\{1,2,3,4,5,6\}$ (3 Marks)

$y = x+1$

$x=1 \Rightarrow y=2$
$x=2 \Rightarrow y=3$
$x=3 \Rightarrow y=4$
$x=4 \Rightarrow y=5$
$x=5 \Rightarrow y=6$
$x=6 \Rightarrow y=7$ but $y=7$ not included in set A

Domain $=(1,2,3,4,5,6)$
Range $=(\cancel{1,2}\ 2,3,4,5,6)$

Q2c $R=\{(2,1)(3,1)(4,1)(3,2)(4,2)(4,3)$ ~~(1,2)(1,3)(1,4)(2,3)(2,4)(3,4)~~ $(1,1)(2,2)(3,3)(4,4)\}$

R is Reflexive yes

R is Antisymm yes

R is Transitive yes

R is Partial order relation

Q2d b/c diagonal should be 1,1,1 for Reflexive so not Equivalence

or

$R=\{(1,a)(1,b)(1,c)(2,c)(3,a)(3,b)(3,c)\}$
Not Reflexive by checking the condition $(a,a)\notin R$